

Quantum CMAB Evaluation Report v5.0

This report presents a comprehensive empirical evaluation of contextual multi-armed bandit models (CMAB) for adversarial quantum routing environments. We evaluated 11 algorithms under stochastic network failures, with CPursuit and EXPNeuralUCB emerging as top performers. Our analysis reveals key weaknesses in early hybrid integration attempts (CExpNeuralUCB) and provides a baseline selection strategy for hybrid models integrating iCMAB forecasting. Recommendations are provided for robust adversarial hybrid development.

Key Findings:

- ✓■ Top Performer: CPursuit (96.4% Oracle efficiency)
- ✓■ Neural Cluster: EXPNeuralUCB (92.1%), GNeuralUCB (90.2%)
- Hybrid Underperformance: CExpNeuralUCB needs refinement (avg 29.2%)
- Multi-run validation: Statistical consistency across 3, 5, 8, and 10 runs
- Framework: Fully reproducible, Colab-ready, adversarial environment pending

Recommended Baseline:

We recommend baselining future hybrid models using CPursuit for statistical reliability and EXPNeuralUCB for adversarial resilience. iCMAB integration should include context-aware gating logic and adaptive feedback.

Next Steps for Hybrid Framework:

1. Refactor CExpNeuralUCB with switchable neural experts.
2. Finalize hybrid benchmark against iCMAB-enhanced adversarial environment.
3. Automate test suite for Tier 1–Tier 4 performance comparison across environments.
4. Deploy QuantumExperimentRunner with hybrid evaluation mode enabled.

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